

A Total Dietary Program Emphasizing Magnesium Instead of Calcium

Effect on the Mineral Density of Calcaneous Bone in Postmenopausal Women on Hormonal Therapy

Guy E. Abraham, M.D.
Harinder Grewal, M.D.

The use of calcium supplementation for the management of primary postmenopausal osteoporosis (PPMO) has increased significantly in the past few years. A review of the published data does not support calcium megadosing during postmenopause. Controlled studies showed no significant effect of calcium intake on mineral density of trabecular bone and a slight effect on cortical bone. Since PPMO is predominantly due to demineralization of trabecular bone, there is no justification for calcium megadosing in postmenopausal women. Soft tissue calcification is a serious risk factor during calcium megadosing under certain conditions. A total dietary program emphasizing magnesium instead of calcium for the management of PPMO takes into account the available data on the effects of magnesium, life-style and dietary habits on bone

integrity and PPMO. When this dietary program was tested on 19 postmenopausal women on hormonal replacement therapy who were compared to 7 control postmenopausal women, a significant increase in mineral bone density of the calcaneous bone (BMD) was observed within one year. Fifteen of the 19 women had had BMD below the spine fracture threshold before treatment; within one year, only 7 of them still had BMD values below that threshold.

Introduction

Osteoporosis is a clinical syndrome characterized by a high susceptibility to bone fracture with resultant symptomatology and is due to an excessive amount of bone loss without any change in bone structure.^{1,2} It is called "primary osteoporosis" when there are no known factors and includes senile osteoporosis, with both cortical and trabecular bone loss, and postmenopausal osteoporosis (PPMO), with mainly trabecular bone loss during the first decade after menopause.^{3,4}

During the last National Institutes of Health workshop on osteoporosis, in 1987, the panel of experts recommended a daily intake of 1,500 mg of calcium by postmenopausal women as a means of preventing PPMO⁵ even though the clinical studies presented in the same proceedings reached different conclusions^{6,7}: calcium supplementation in daily amounts of $\leq 3,000$ mg had no significant effect on postmenopausal trabecular bone loss above the placebo effect.

One of us (G.E.A.) previously postulated that PPMO is a skeletal manifestation of magnesium deficiency with the corollary that postmenopausal trabecular bone loss would not occur even without estrogen therapy if the magnesium intake were sufficient to maintain an adequate bone magnesium reserve.⁸ That author proposed a total dietary program emphasizing magnesium instead of calcium, not unlike the dietary recommendations for women with the premenstrual tension syndrome.⁹

To test the above postulate, the magnesium-emphasized program was implemented for 6–12 months in 19 postmenopausal women receiving hormonal replacement therapy. Seven postmenopausal women on hormonal replacement therapy served as controls. Trabecular bone density was assessed with single photon densitometry of the calcaneous bone.¹⁰

Materials and Methods

Twenty-six postmenopausal women were recruited from a menopause clinic. All subjects were on hormonal replacement therapy, either estrogen alone in those with surgical menopause or cyclic progestogen

From Optimox, Inc., Torrance, and Women's Life Care, Anaheim Hills, California.

Address reprint requests to: Guy E. Abraham, M.D., 2720 Monterey Street, Suite 406, Torrance, CA 90503.

superimposed on estrogen therapy in those with an intact uterus.⁷

All patients underwent bone density measurement of the calcaneous bone (BMD) with single photon absorptiometry, as described by Ross et al,¹⁰ and were advised about the dietary program (Figure 1). Micronutrients were supplied in the form of a complete multivitamin, multimineral supplement (Gynovite, Optimox, Inc., Torrance, CA) containing 500 mg of calcium as the citrate salt and 200 mg of magnesium as the oxide (Table I). Seven patients received dietary advice but chose not to take the supplement. Nineteen patients received dietary advice and ingested six tablets of the nutritional supplement daily. Besides the supplement, magnesium oxide was given at a daily dosage of 400 mg of elemental magnesium. Therefore, the 19 patients received 50% of the recommended daily allowance (RDA) of calcium and 200% of the RDA of magnesium for women. In all 26 patients, BMD of the calcaneous bone was repeated at their return visit, 6–12 months later. The radioactive materials were replaced every three months to prevent artifacts from significant radioactive decay, resulting in a prolonged counting time during BMD measurement.

Results

Comparing the two groups of patients, there was no significant difference in age, height, weight, years since menopause, duration of hormone therapy, baseline BMD or duration of follow-up (Table II).

A nonsignificant increase, 0.7%, in the mean BMD of the seven patients receiving hormonal therapy and dietary advice was observed as compared to a mean

1. Limit intake of protein to 1.5 g/kg body weight/d. Consume twice as much vegetable protein as animal protein.
2. Rely more on fish, poultry, whole grains and legumes as a source of protein and less on red meat and dairy products.
3. Limit the intake of fats to 30% of the total calories. Favor vegetable oils over cooked fats and animal fats.
4. Increase the intake of complex carbohydrates (45–60% of the total calories) from cereals, whole grains and legumes.
5. Limit the consumption of refined sugar (5 tsp/d), salt (3 g/d), red meat (3 oz/d), alcohol (1 oz/d), coffee, tea and chocolate.
6. Limit tobacco use.
7. Limit the intake of dairy products to two servings per day.
8. Avoid processed foods.

Figure 1

Summary of dietary recommendations for postmenopausal women.

Table I Composition of Six Tablets of Gynovite

Composition	Amount
Liposoluble vitamins	
Vitamin A (palmitate) (water dispersible)	5,000 IU
Vitamin E (d- α tocopherol acid succinate)	400 IU
Vitamin D ₃ (cholecalciferol)	400 IU
Hydrosoluble vitamins (sustained release)	
Folic acid	400 μ g
Vitamin B ₁ (thiamine mono)	10 mg
Vitamin B ₂ (riboflavin)	10 mg
Niacinamide	20 mg
Vitamin B ₆ (pyridoxine HCl)	20 mg
Vitamin B _{12a}	125 μ g
Pantothenic acid (d-calcium pantothenate)	10 mg
Vitamin C (ascorbic acid)	180 mg
Minerals	
Calcium (citrate)	500 mg
Magnesium (oxide)	200 mg
Iodine (hydrolyzed protein complex)	150 μ g
Iron (amino acid chelate)	18 mg
Copper (amino acid chelate)	2.0 mg
Zinc (amino acid chelate)	15 mg
Manganese (amino acid chelate)	10 mg
Selenium (hydrolyzed protein complex)	200 μ g
Chromium (hydrolyzed protein complex)	200 μ g
Other ingredients (in descending order of weight)	
Boron (hydrolyzed protein complex)	
Betaine acid HCl	
Pancreatin	
Hesperidin	
Para amino benzoic acid	
Inositol	
Rutin	
Biotin	

increase of 11% in the 19 women receiving the supplements ($P < .01$ by paired data analysis) (Table II).

Discussion

Since several nutrients besides calcium are important for bone integrity,^{11–24} a total dietary program would be preferable to calcium supplementation in the prevention and management of PPMO.

Albanese et al¹¹ reported that calcium supplementation alone was two to three times less effective on phalanx bone density in women aged 38–66 years than was nutritional supplementation containing the same amount of calcium plus some essential trace elements and vitamins at RDA levels. However, Albanese's supplement lacked several essential micronutrients and did not contain magnesium.

The results of the present study suggest that complete nutritional supplementation containing 500 mg of calcium as the citrate salt and 600 mg of magnesium as the oxide has a significant effect on reversing

Table II Effect of the Nutritional Supplement on the Density of Calcaneous Bone in Postmenopausal Women

No.	Age (yr)	Height (in)	Weight (lb)	Age at menopause	Bone density (g/cm ²)		Time interval (mo)
					Before	After	
Women not taking supplements							
1	42	65	138	39 S	0.307	0.294	9
2	42	61	121	36 S	0.315	0.330	9
3	72	63	105	49 N	0.226	0.225	7
4	51	62	108	33 N	0.276	0.256	12
5	58	63	113	43 S	0.314	0.320	12
6	49	63	145	45 S	0.343	0.356	12
7	49	69	199	43 S	0.339	0.353	8
Mean ±SD	52 ±9.7	63.7 ±2.6	132.7 ±32.8	41.1 ±5.4	0.303 ±0.041	0.305 ±0.050	9.8 ±2.1
Women taking supplements							
1	58	65	135	32 S	0.210	0.221	12
2	55	67	132	37 S	0.302	0.306	8
3	53	65	135	31 S	0.317	0.321	12
4	49	62	142	37 S	0.272	0.345	8
5	48	60	106	48 N	0.296	0.428	9
6	45	62	140	31 N	0.313	0.358	6
7	49	67	132	49 N	0.336	0.326	6
8	55	60	128	50 S	0.319	0.369	6
9	61	63	130	50 N	0.312	0.320	12
10	28	63	122	23 S	0.454	0.458	12
11	53	64	142	40 S	0.339	0.376	7
12	75	62	120	31 N	0.284	0.294	9
13	66	65	105	49 N	0.299	0.286	12
14	45	63	122	43 N	0.307	0.316	12
15	52	64	130	47 S	0.265	0.292	8
16	70	63	125	36 S	0.253	0.341	6
17	51	69	105	45 S	0.296	0.303	9
18	52	65	188	40 S	0.376	0.386	7
19	49	65	145	49 N	0.314	0.363	9
Mean ±SD	53 ±10	63.7 ±2.2	130.6 ±17.9	40.3 ±8.0	0.303 ±0.048	0.337 ±0.052	8.5 ±2.9

N = normal menopause.

S = surgical menopause.

All subjects were on hormonal replacement therapy before and during the study.

postmenopausal bone loss of the calcaneous bone within a relatively short period of time in patients receiving hormonal replacement therapy for several years. The effect of this magnesium-emphasized program on calcaneous bone density was 16 times greater than that of dietary advice alone in postmenopausal women on hormonal replacement therapy. Ross et al¹⁰ defines the spine fracture threshold as a BMD of 0.32 g/cm² of the calcaneous bone. In 15 of the 19 women receiving the supplement the BMD was below the fracture threshold before treatment. Within a year after the program only seven patients had BMD values < 0.32 g/cm² (Table II).

Dalderup was the first to report, in 1960, a possible role of magnesium in therapy for osteoporosis.²⁵ He also warned against soft tissue calcification caused by calcium and vitamin D megadosing.

A recent literature review suggested that the magnesium content of the food supply in Europe and North America results in a daily intake averaging 72–161 mg below women's RDA of 300 mg.²⁶ Magnesium supplementation ranging from 67 to 600 mg daily improved several clinical problems in those countries.

The RDA of magnesium for Soviet women varies from 500 to 1250 mg, depending on physiologic conditions.²⁷ It is unlikely that genetic factors account for such a difference in magnesium requirements between Soviet and U.S. women. The U.S. RDA of magnesium, based on short-term balance studies, probably is the minimum daily intake of magnesium that the human body can adjust to²⁸ but at the cost of increased susceptibility to stress²⁹ and, very probably, PPMO.⁸ Long-term balance studies have indicated

much greater needs for magnesium than the U.S. RDA, and >1,000 mg/d is sometimes required to maintain a positive balance under stressful conditions.²⁸

It was postulated by one of us (G.E.A.) that PPMO is predominantly a skeletal manifestation of chronic magnesium deficiency, facilitated by estrogen withdrawal during postmenopause. If this postulate is correct and properly designed clinical trials could test it easily, postmenopausal trabecular bone loss might not occur even without estrogen replacement therapy as long as the magnesium intake and bone magnesium reserve were adequate. Even senile osteoporosis would be prevented by this program because the high-magnesium diet would lower the calcium threshold.⁸ Raising the RDA of magnesium to 1,000 mg/d and lowering that for calcium to 500 mg/d might be the most cost-effective approach to PPMO at a national level. This proposed RDA for calcium would be more in line with the World Health Organization's "practical allowance" of 400–500 mg daily for adults.³⁰ Such a reversal of the magnesium:calcium ratio in the RDA recommendation most probably would lower the incidence and prevalence of many other degenerative diseases and pregnancy complications caused in part by magnesium deficiency.^{26,31}

We are now undertaking a more comprehensive study of the magnesium-emphasized dietary program in postmenopausal women not receiving hormonal replacement because of contraindications to estrogen use. The daily supplementation of elemental magnesium will vary from 200 to 1,000 mg. If the above postulate is valid, BMD changes would be expected to correlate positively with the amount of magnesium ingested.

If, indeed, a magnesium-emphasized dietary program reverses bone loss in PPMO, this program for PPMO might be the most cost-effective one and essentially devoid of side effects.

References

1. National Institutes of Health Consensus Development Conference: Osteoporosis. *JAMA* 252:799, 1984
2. National Institutes of Health Consensus Development Conference on Osteoporosis, Bethesda, April 1984
3. Riggs BL, Melton LJ III: Heterogeneity of involutional osteoporosis: Evidence for two osteoporosis syndromes. *Am J Med* 75:899, 1983
4. Johnston CC, Norton J, Khairi MRA, et al: Heterogeneity of fracture syndromes in postmenopausal women. *J Clin Endocrinol Metab* 61:551, 1985
5. Chesnut CH: Report from the NIH Consensus Conference, 1984, and NIH/NOF Workshop, 1987. *In* Osteoporosis Update, 1987. Edited by HK Genant. San Francisco, Radiology Research and Education Foundation, 1987, pp 3–6
6. Ettinger B: Estrogen, progestogen, and calcium in treatment of postmenopausal women. *In* Osteoporosis Update, 1987. Edited by HK Genant. San Francisco, Radiology Research and Education Foundation, 1987, pp 253–58
7. Christiansen C, Riis BJ: Optimal prophylaxis for postmenopausal bone loss. *In* Osteoporosis Update, 1987. Edited by HK Genant. San Francisco, Radiology Research Education Foundation, 1987, pp 259–266
8. Abraham GE: The calcium controversy. *J Appl Nutr* 34:69, 1982
9. Abraham GE: Nutritional factors in the etiology of the premenstrual tension syndromes. *J Reprod Med* 28:446, 1983
10. Ross PD, Wasnich RD, Heilbrun LK, et al: Definition of a spine fracture threshold based upon prospective fracture risk. *Bone* 8:271, 1987
11. Albanese AA, Lorenze EJ, Edelson HA, et al: Calcium nutrition and skeletal bone health. *Nutr Rep Int* 38:211, 1988
12. Barischoe AM, Ragen C: Relation of magnesium on calcium metabolism in man. *Am J Clin Nutr* 19:296, 1966
13. Benke PJ, Fleshood HL, Pitot HC: Osteoporotic bone disease in the pyridoxine-deficient rat. *Biochemical Med* 6:526, 1972
14. Bikle DD, Murphy EW, Rasmussen H: The ionic control 1.25-dihydroxyvitamin D₃ synthesis in isolated chick renal mitochondria: The role of potassium. *Biochem Biophys Acta* 437:394, 1976
15. Carlisle EM: Silicon as an essential trace element in animal nutrition. *In* Silicon Biochemistry. Edited by ED O'Connor. New York, John Wiley & Sons, 1986, pp 123–39
16. Cohen L, Kitzes R: Infrared spectroscopy and magnesium content of bone mineral in osteoporotic women. *Isr J Med Sci* 17:1123, 1981
17. Freudenheim JL, Johnson NE, Smith EL: Relationships between usual nutrient intake and bone-mineral content of women 35–65 years of age: Longitudinal and cross-sectional analysis. *Am J Clin Nutr* 44:863, 1986
18. Medalle R, Waterhouse C, Hahn TJ: Vitamin D resistance in magnesium deficiency. *Am J Clin Nutr* 29:858, 1976
19. Mirra JM, Alcock NW, Shils ME: Effects of calcium and magnesium deficiencies on rat skeletal development and parathyroid gland area. *Magnesium* 1:16, 1982
20. Nielsen FH, Hunt CD, Mullen LM: Effect of boron on mineral, estrogen, and testosterone metabolism in postmenopausal women. *Fed Am Soc Exp Biol J* 1:394, 1987
21. Okazaki M: Magnesium action on the stability of fluorapatite. *Magnesium* 7:148, 1988
22. Okazaki M: Mg²⁺-F-interaction during hydroxyapatite formation. *Magnesium* 6:296, 1987
23. Strause LG, Hegenauer J, Saltman P: Effects of long-term dietary manganese and copper deficiency on rat skeleton. *J Nutr* 116:135, 1986
24. Yano K, Heilbrun LK, Wasnich RD, et al: The relationship between diet and bone mineral content of multiple skeletal sites in elderly Japanese-American men and women living in Hawaii. *Am J Clin Nutr* 42:877, 1985

25. Dalderup LM: The role of magnesium in osteoporosis and idiopathic hypercalcaemia. *Voeding* 21:424, 1960
26. Marier JR: Magnesium content of the food supply in the modern-day world. *Magnesium* 5:1, 1986
27. Lederer J: *Magnesium: Mythes et réalité*. Paris, Maloine Editeurs, 1984, p 54
28. Seelig MS: Magnesium requirements in human nutrition. *Magnesium Bull* 1a:26, 1981
29. Heroux O, Peter D, Heggtevit HA: Long-term effect of suboptimal dietary magnesium. *J Nutr* 107:1640, 1977
30. Kanis JA, Passmore R: Calcium supplementation of the diet: I and II. *Br Med J* 298:137, 205, 1989
31. Seelig MS: *Magnesium Deficiency in the Pathogenesis of Disease*. New York, Plenum Medical Book Company, 1980